

DERIVATIVES II: OPTIONS AND THE GREEKS

(DEFINITION, TYPES, MONEYNES, PAYOFF AND GAIN FUNCTIONS, PRICING,
RISK TYPES)

DEFINITION

An **option** gives the *right but not the obligation* to the option holder to exercise the contract. By contrast, in the case of a forward or futures contract the two parties have committed themselves to some action. It costs a trader nothing to enter into a forward or futures contract, while *there is an up-front payment in the case of an option*. Options are attractive for two reasons: they **provide leverage** (higher upside potential) and **provide insurance** (downside is limited). A call (put) option should be worth more than a similar long (short) forward, bc it contains no obligation.

TYPES OF OPTIONS

Call option: Gives the holder the *right, but not the obligation to buy the underlying asset* at a specified strike/exercise price at a specified expiration date/maturity date.

Put option: Gives the holder the *right, but not the obligation to sell the underlying asset* at a specified strike/exercise price at a specified expiration date/maturity date.

Option positions

There are two sides to every option contract. On one side is the investor who has taken the *long position* (i.e., has bought the option). On the other side is the party who has taken a *short position* (i.e., has sold or written the option). The writer (seller) of the option receives cash up-front but has potential liabilities later. The writer's profit/loss is the reverse of that for the purchaser of the option.

There are four types of option positions ($S(T)$ is underlying's price at maturity, K = strike price):

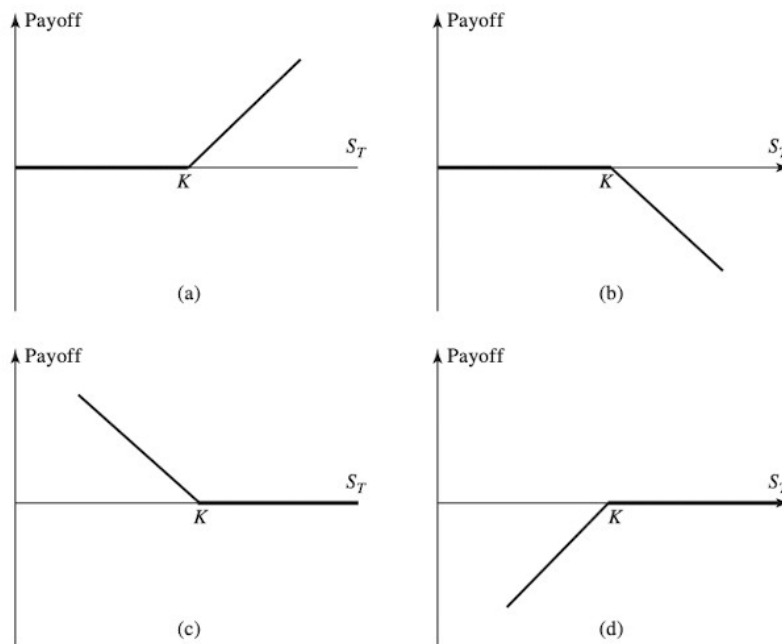
	LONG	SHORT
CALL	$\max(S_T - K, 0)$	$\min(K - S_T, 0)$
PUT	$\max(K - S_T, 0)$	$\min(S_T - K, 0)$

When calculating the *payoffs* (payoff function) to the purchaser of the option the initial cost of the option is not included in the calculation.

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Figure 10.5 Payoffs from positions in European options: (a) long call; (b) short call; (c) long put; (d) short put. Strike price = K ; price of asset at maturity = S_T .



Types of underlying assets

- *Stock Options*: One contract gives the holder the right to buy or sell 100 shares at a specified strike price.
- *ETP (Exchange Traded Products) Options*: ETPs are listed on exchanges and traded like stocks. e.g. ETFs.
- *Foreign Currency Options*: Mostly on the over-the-counter market, and some exchange trading. One contract gives the right to buy or sell 10,000 units of a foreign currency for U.S. dollars.
- *Index Options*: Contract is usually for 100 times the index at the specified strike price, settlement is always in cash, rather than delivering the portfolio underlying the index (e.g. S&P 500 Index)
- *Futures Options*: Life of a futures option normally ends a short period of time before the expiration of trading in the underlying futures contract. When a call option is exercised, the holder's gain equals the excess of the futures price over the strike price.

American vs European options

An American option allows its holder to exercise the right to purchase (if a call) or sell (if a put) the underlying asset *on or before the expiration date*. European options allow for exercise *only on the expiration date*. American-style options, because they allow more optionality than their

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European counterparts, generally will be more valuable. It is never worth to exercise an American call on a non-dividend paying stock.

Summary of variables affecting stock option prices:

<i>Variable</i>	<i>European call</i>	<i>European put</i>	<i>American call</i>	<i>American put</i>
Current stock price	+	–	+	–
Strike price	–	+	–	+
Time to expiration	?	?	+	+
Volatility	+	+	+	+
Risk-free rate	+	–	+	–
Amount of future dividends	–	+	–	+

+ indicates that an increase in the variable causes the option price to increase or stay the same;

– indicates that an increase in the variable causes the option price to decrease or stay the same;

? indicates that the relationship is uncertain.

Call (put) options become more (less) valuable as stock price increases but become less (more) valuable as strike price increases. Increase in volatility leads in value increase for both european and american options.

Assumptions when considering option pricing:

- 1) No transaction costs.
- 2) All trading profits (or losses) are the subject of the same tax rate.
- 3) Borrowing and lending are possible at the risk-free interest rate.

Exotic options

- **Bermudan Options**

Can be exercised *on specific dates* before expiration, not just anytime (like American options) or only at expiration (like European options). It's used in reducing the impact of price manipulation or volatility.

- **Asian options**

Options with payoffs that depend on the average price of the underlying asset during at least some portion of the life of the option.

- **Barrier options**

Options with payoffs that depend not only on some asset price at option expiration, but also on whether the underlying asset price has crossed through some “barrier”. These options activate

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(knock-in) or deactivate (knock-out) when the underlying asset reaches a certain price level (the barrier).

- **Binary Options**

They pay a fixed amount if a certain condition is met; otherwise, they pay nothing.

MONEYNES

- *ATM (at the money)*: has strike price equal to the current market price.
- *ITM (in the money)*: has strike price favorable compared to the market price, meaning positive cash flow.
- *OTM (out of the money)*: has strike price unfavorable compared to the market price, meaning negative cash flow.

PAYOFF AND GAIN FUNCTIONS

Boundaries for Option Prices

If an option price is above the upper bound or below the lower bound, then there is an arbitrage opportunity.

	European		American	
	Lower Bound	Upper Bound	Lower Bound	Upper Bound
Call	$\max(0; S(T) - PV(K))$	$S(0)$	$\max(0; S(T) - PV(K))$	$S(0)$
Put	$\max(0; PV(K) - S(T))$	$PV(K)$	$\max(0; K - S(T))$	K

Upper bound rules:

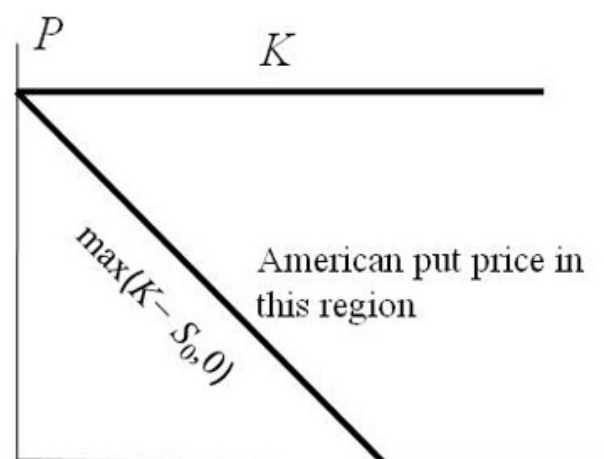
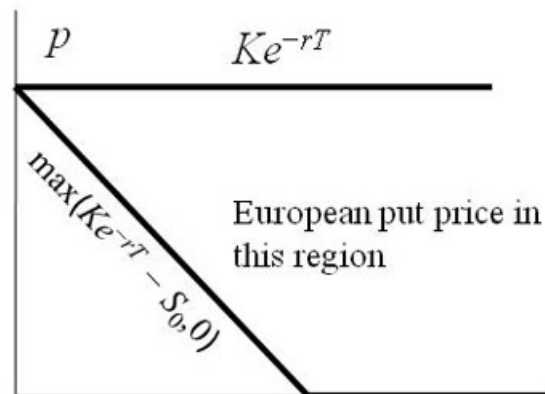
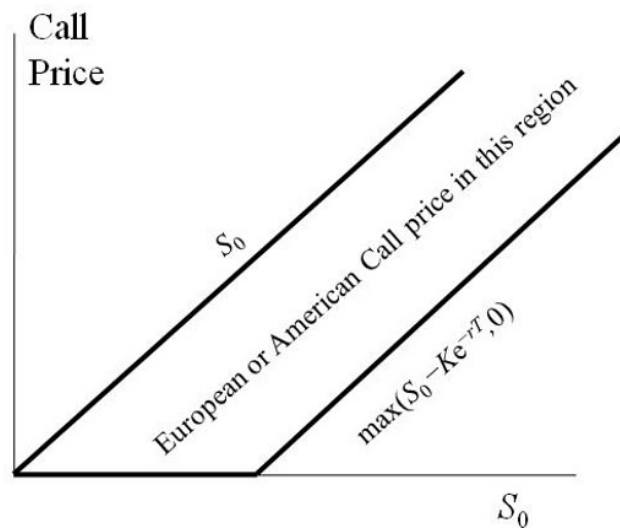
- EU/US call options: option can never be worth more than the stock.
- US put options: no matter how low the stock price goes the option can't be worth more than K (it's worth K if $S=0$).
- EU put option: At maturity the option can't be worth more than K , so it can't be worth more than the present value of K .

Lower bound rules:

- It's a right but not an obligation to exercise the option, therefore option value cannot be negative.
- EU call: $S_0 - PV(K)$
- EU put: $PV(K) - S_0$

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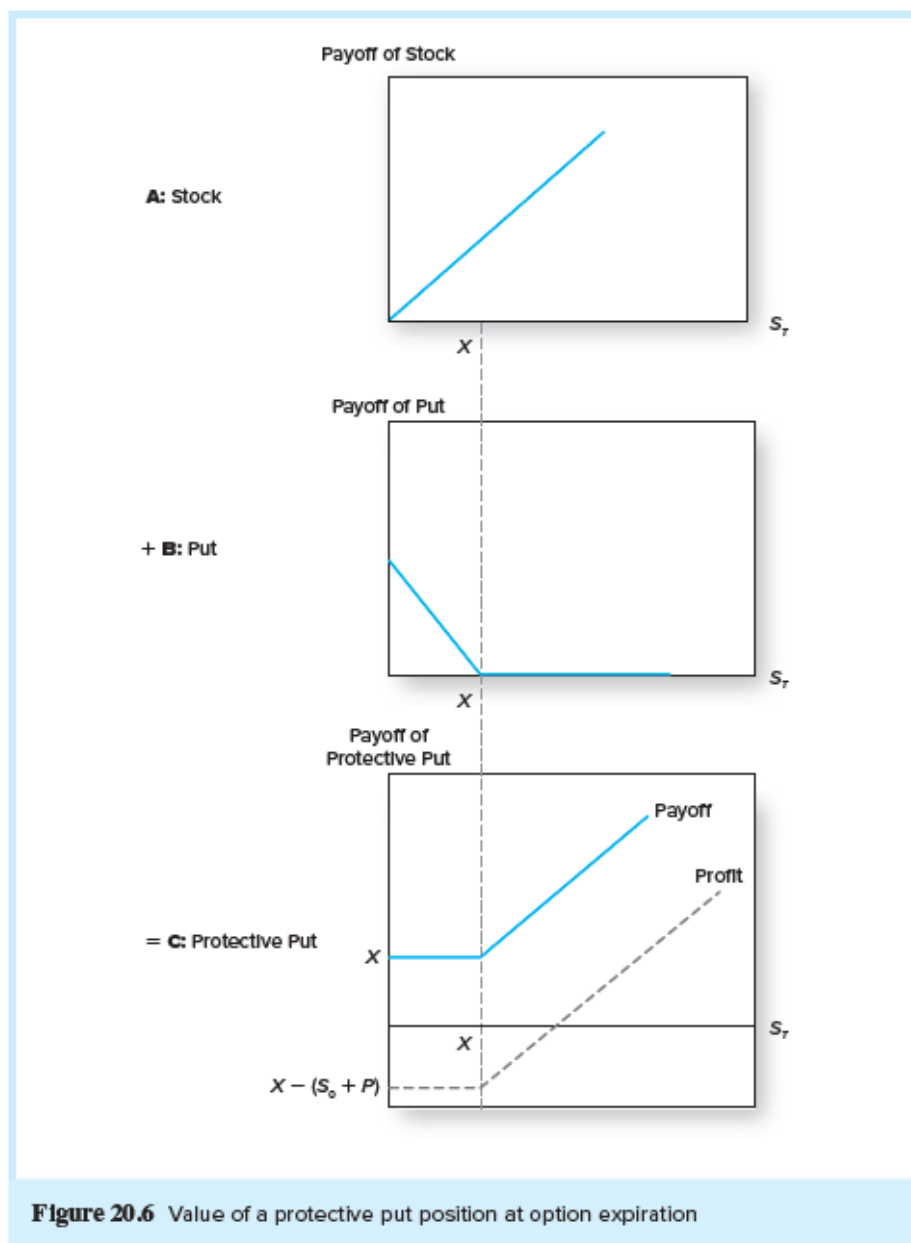
Trading Strategies with Options

- **Protective Put**

Consists of *investing in the stock and long put option* on the stock. The payoff will be at least the put option's exercise price. The strategy limits losses so they can provide a form of portfolio insurance.

	$S_T \leq X$	$S_T > X$
Stock	S_T	S_T
+ Put	$X - S_T$	0
Total	X	S_T

The payoff and profit is the following:



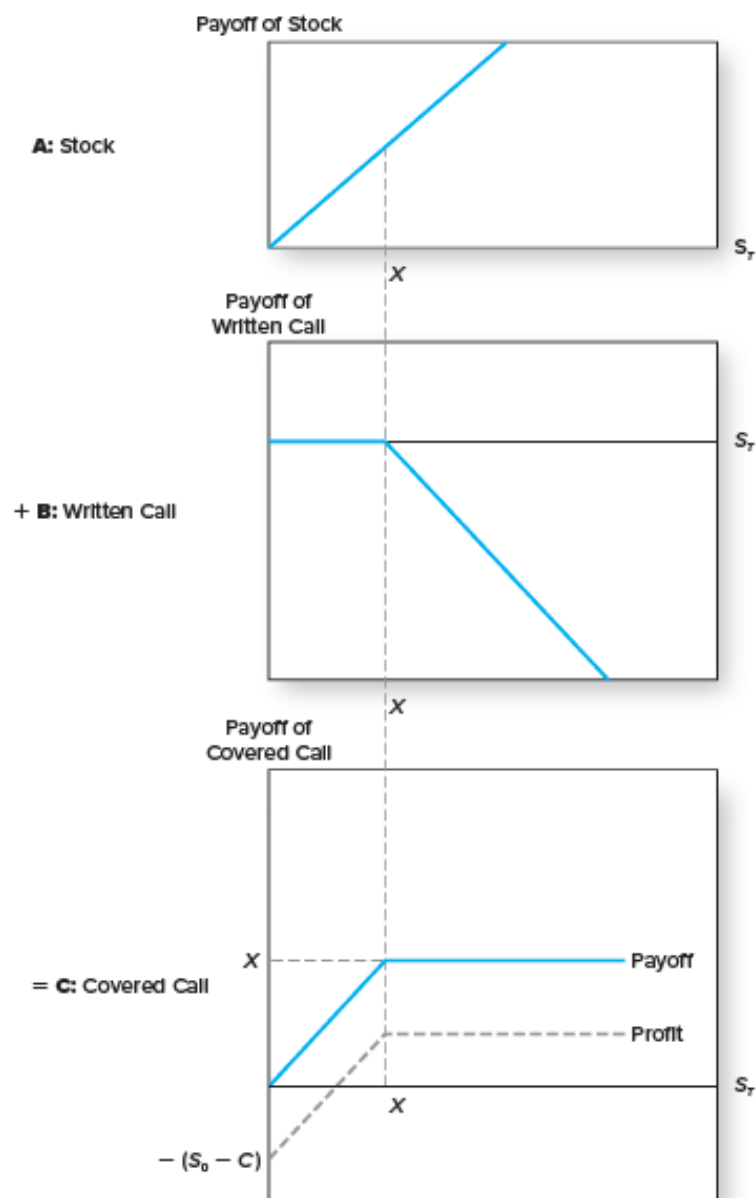
- **Covered call**

The **purchase of a stock and a short call** on the stock. The obligation is covered with the bought stock if the option is exercised.

Table 20.2

Value of a covered call position at option expiration

	$S_T \leq X$	$S_T > X$
Payoff of stock	S_T	S_T
+ Payoff of written call	-0	$-(S_T - X)$
Total	S_T	X

**Figure 20.8** Value of a covered call position at expiration

• Spreads

A spread is a **combination of two or more call options** (or two or more puts) on the same stock with differing exercise prices or times to maturity.

Bull Spread

- with puts: $LP(K_1) + SP(K_2)$, $K_1 < K_2$
- with calls: $LC(K_1) + SC(K_2)$, $K_1 < K_2$

Figure 12.2 Profit from bull spread created using call options.

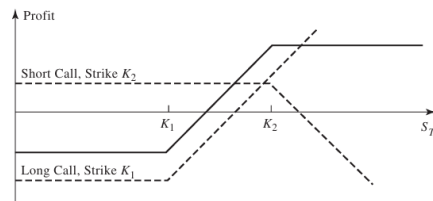
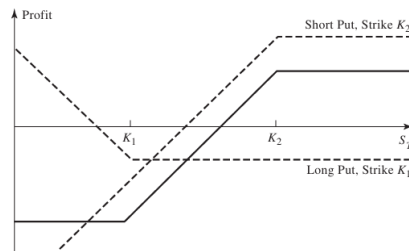


Figure 12.3 Profit from bull spread created using put options.



Bear Spread

- With puts: $LP(K_2) + SP(K_1)$; $K_2 > K_1$
- With calls: $LC(K_2) + SC(K_1)$; $K_2 > K_1$

Figure 12.4 Profit from bear spread created using put options.

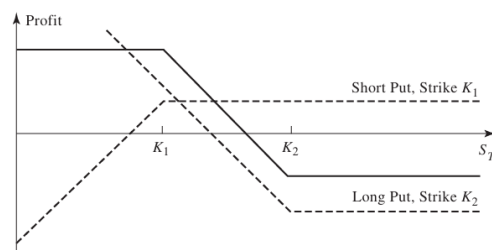
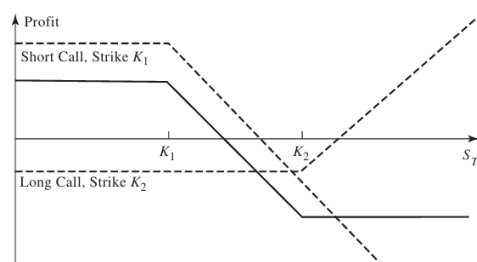


Figure 12.5 Profit from bear spread created using call options.



- **Straddle**

A long straddle is established by **buying both a call and a put**, each with the same exercise price, X , and the same expiration date, T . The worst-case scenario in this strategy is no movement in the stock price. It is a *bet on volatility*.

Long Straddle = Long Call + Long Put with the same Strike price.

Short Straddle = Short Call + Short Put with the same Strike price.

Table 20.3

Value of a straddle position at option expiration

	$S_T < X$	$S_T \geq X$
Payoff of call	0	$S_T - X$
+ Payoff of put	$X - S_T$	0
Total	$X - S_T$	$S_T - X$

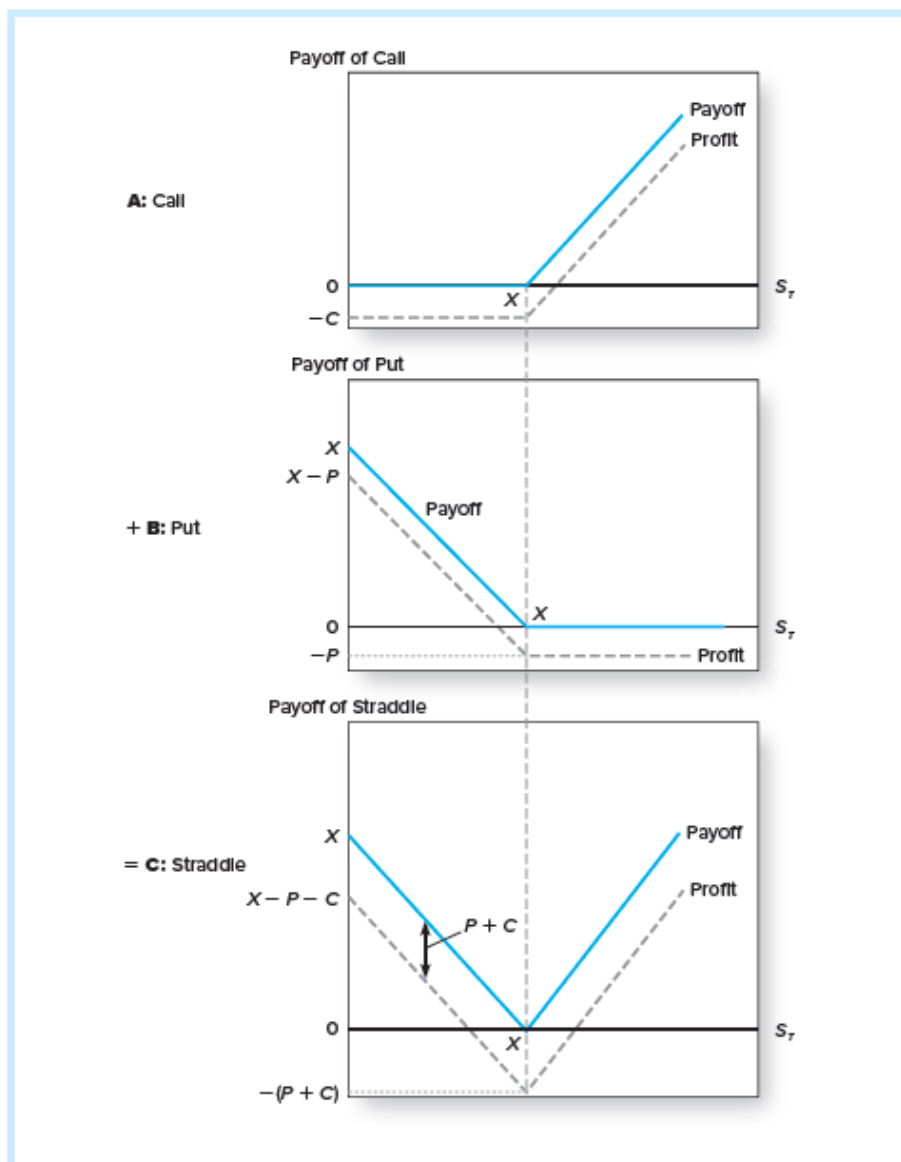


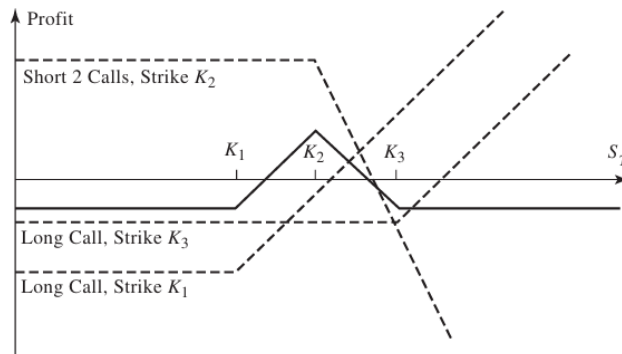
Figure 20.9 Value of a straddle at expiration

- **Butterfly**

A butterfly spread involves positions in options with three different strike prices. It can be created by buying a European call option with a relatively low strike price K_1 , buying a European call option with a relatively high strike price K_3 , and selling two European call options with a strike price K_2 that is halfway between K_1 and K_3 .

Long Butterfly with calls = $LC(K_1) + 2 * SC(K_2) + LC(K_3)$

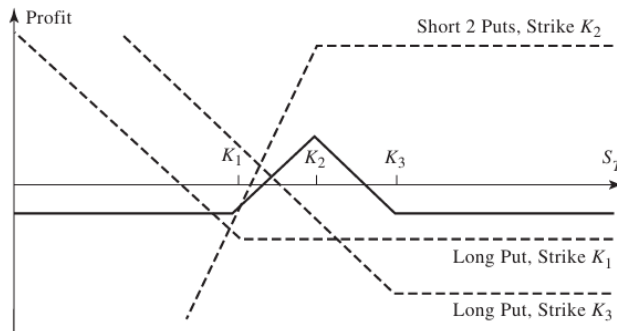
Figure 12.6 Profit from butterfly spread using call options.



The investor buys two European puts, one with a low strike price and one with a high strike price and sells two European puts with an intermediate strike price.

Long Butterfly with puts = $LP(K_1) + 2 * SP(K_2) + LP(K_3)$

Figure 12.7 Profit from butterfly spread using put options.

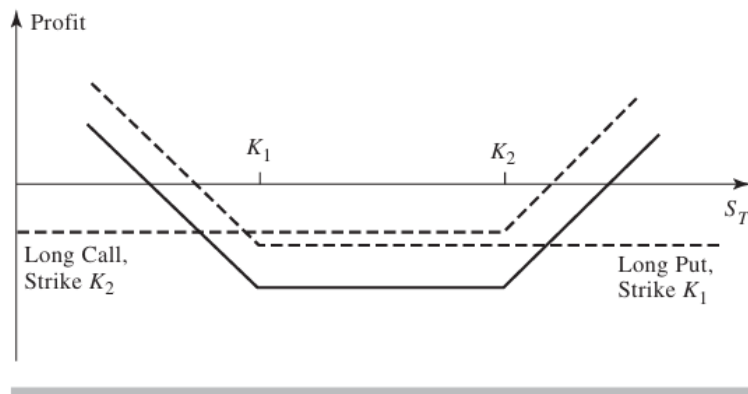


A butterfly spread can be sold or shorted by following the reverse strategy. Options are sold with strike prices of K_1 and K_3 , and two options with the middle strike price K_2 are purchased. This strategy produces a modest profit if there is a significant movement in the stock price.

- **Strangle**

In a strangle an investor buys a European put and a European call with the same expiration date and different strike prices. The investor is betting that there will be a large price move but is uncertain whether it will be an increase or a decrease.

Figure 12.12 Profit from a strangle.



Parities on the markets

We can purchase an underlying asset in several ways. Purchases can occur on the *spot*, *futures* or *options market*. If we buy on the spot market we pay in the present; if on the futures market, we pay in the future; if on the options market, we pay at both times. However, these payments must be equal regardless of time. If equality of payments does not hold across different markets, arbitrage is possible. Based on this, the following equalities must be held between the markets.

Forward price (between spot and forward market):

$$LF = LU + SB \implies F = S_t(1 + r_f)^{T-t}$$

Put-call parity (between spot and option market): it is the relationship between the price of a European call, European put, and the underlying.

$$LF = LC + \bar{S}P \implies F_T - K = (c - p)(1 + r_f)^{T-t}$$

$$S_t - PV(K) = c - p$$

- European options

$$c + Ke^{-rT} = p + S_0$$

- American options

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$

OPTION PRICING

Risk neutral valuation

Risk neutral valuation states that when valuing a derivative, we can assume that *investors are risk-neutral*: investors do not increase the expected return they require from an investment to compensate for increased risk. Assuming a risk-neutral world gives us the *right option price* for the world we live in, as well as for a risk-neutral world. It solves the problem that we don't know anything about the risk aversion of option investors. When we are pricing an option in terms of the price of the underlying stock, risk preferences are unimportant.

Under risk-neutral probabilities:

- Expected returns on all assets are equal to the risk-free rate.
- The discount rate used for the expected payoff on an option (or any other instrument) is the risk-free rate.
- Asset prices remain consistent with the observed market prices.

The reason they work so well is that market prices already account for:

- Real-world expectations of future outcomes.
- The aggregate risk preferences of market participants.
- External factors like liquidity, economic conditions, and supply/demand for specific options.

Parameter p should be interpreted as the probability of an up movement in a risk-neutral world:

$$p = \frac{e^{rT} - d}{u - d}$$

The value of the option today is its *expected future payoff in a risk-neutral world discounted at the risk-free rate*.

$$E(S_T) = pS_0u + (1 - p)S_0d$$

$$E(S_T) = pS_0(u - d) + S_0d$$

$$E(S_T) = S_0e^{rT}$$

The stock price grows, on average, at the risk-free rate when p is the probability of an up movement.

Steps of risk-neutral valuation:

- Calculate what the probabilities of different outcomes would be if the world were risk-neutral
- Calculate the expected payoff from the derivative
- Discount the expected payoff at the risk-free rate

Valuation methods

The standard procedure for valuing an asset is to (1) figure out *expected cash flows* and (2) discount them at the *opportunity cost of capital*. The first step is messy but feasible; however, finding the opportunity cost of capital is *impossible* because the **risk of an option changes every time the stock price moves**. A deep ITM option is almost certain to be exercised and is therefore safer than a call that is OTM. So, a stock price increase (relative to the strike price) raises the option's expected payoff and reduces its risk.

1.) Replicating portfolio / No-arbitrage argument**a. Stock + borrowing**

Set up an *option equivalent* by a combination of an *investment in common stock and borrowing*. Because they have the same payoffs, the Law of One Price implies that the current value of the call and the replicating portfolio must be equal. If we know the strike price, the up and down values of S , we can calculate the payoff from the option, and with that the Δ and the value of borrowing.

Example:

Consider a six-month call option on Amazon stock with an exercise price of \$1,830. The stock price is also \$1,830, option is ATM. The short-term, risk-free interest rate was $r = 2\%$ for six months, or just over 4% per annum. Amazon stock can do only two things over the option's six-month life. The price could rise by 20% to $1,830 \times 1.20 = \$2,196$ or fall by the same proportion to $1,830 \div 1.20 = \$1,525$ a year. The possible payoffs to the option:

	Stock Price = \$1,525	Stock Price = \$2,196
1 call option	\$0	\$366

Compare these payoffs with what you would get if you bought 0.54545 Amazon share and borrowed the present value of \$831.82 from the bank:

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	Stock Price = \$1,525	Stock Price = \$2,196
0.54545 share	\$831.82	\$1,197.82
Repayment of loan + interest	<u>-831.82</u>	<u>-831.82</u>
Total payoff	\$ 0	\$ 366.00

The payoffs from the levered investment in the stock are identical to the payoffs from the call option.

$$\begin{aligned}\text{Value of call} &= \text{value of 0.54545 shares} - \text{value of bank loan} \\ &= 0.54545 \times \$1,830 - 831.82/1.02 = \$998.18 - \$815.51 = \$182.67\end{aligned}$$

The number of shares needed to replicate one call is called the **hedge ratio** or **option delta (Δ)**.

$$\Delta = \frac{C_u - C_d}{S_u - S_d} \quad \text{and} \quad B = \frac{C_d - S_d \Delta}{1 + r_f}$$

B is negative: borrowing is equivalent to a negative investment in the risk-free asset. The replicating portfolio consisted of an investment of just under \$1,000 in the share, which we largely financed by a loan of about \$815. Thus, a call option on a share is equivalent to financing the purchase of the share with borrowed money. That is why the *call option is always riskier than the share* and why we say that call options create leverage.

Option Price in the Binomial Model

$$C = S\Delta + B$$

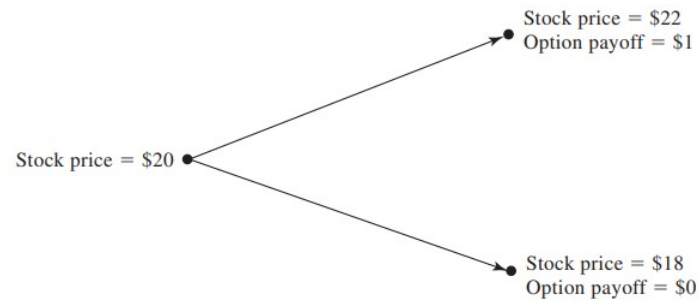
While the *option and the replicating portfolio* do not have the same payoffs in general, they *have the same payoffs given the only two outcomes* we have assumed possible for the stock price. By using the Law of One Price, we are able to solve for the price of the option *without knowing the probabilities of the states* in the binomial tree. This means that ***we do not need to know the expected return of the stock***, which will depend on these probabilities.

b. Stock + option

We assume that there is ***no arbitrage opportunity***, and we set up a *portfolio (by choosing Δ) of the stock and the option* so that there is *no uncertainty* about the value of the portfolio at the end of the 3 months. Because the portfolio has no risk, the return it earns must equal the ***risk-free interest rate***. So the portfolio consists of **Δ long shares** and a **short call option**.

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The strike price is \$21. We want the final portfolio value to be the same for the two alternatives:

$$22\Delta - 1 = 18\Delta \Rightarrow \Delta = 0.25$$

Since we know the final value for sure, there is no risk. So the riskless portfolio consists of 0.25 shares and 1 short call option, and the value of the portfolio will be 4.5 regardless of the stock price's move:

$$22 * 0.25 - 1 = 18 * 0.25 = 4.5$$

The value of the portfolio today is (since it's riskless):

$$4.5 * e^{-0.04 * 0.25} = 4.455$$

If the stock price today is \$20, then the option price c is:

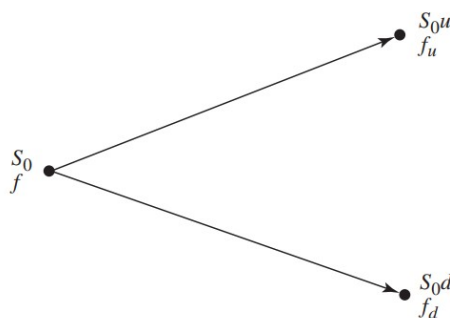
$$20 * 0.25 - c = 4.455 \Rightarrow c = 0.545$$

$$c = 0.545$$

Stock price movement probability:

- It's irrelevant because we are not valuing the option in absolute terms but calculating in terms of the price of the underlying, and the probabilities of up or down movements are already incorporated in the stock price.

If the no-arbitrage assumption holds, then:



$$S_0 u \Delta - f_u = S_0 d \Delta - f_d$$

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

$$f = e^{-rT} [p f_u + (1 - p) f_d]$$

2.) Binomial Option Pricing Model

This model is more flexible than the BSM model and can be used to price both European and American options. It models the underlying asset's price movement over discrete time steps, allowing for variable volatility, dividend payments, and early exercise features.

Option value at a node is:

$$f = e^{-rT} [p f_u + (1 - p) f_d]$$

In a two-step binomial tree the variables p^2 , $2p(1 - p)$, and $(1 - p)^2$ are the probabilities that the upper, middle, and lower final nodes will be reached.

American option

The value of the option at the final nodes is the same as for the European option. At earlier nodes the value of the option is the greater of:

1. $f = e^{-r\Delta t} [p f_u + (1 - p) f_d]$
2. The payoff from early exercise.

Parameters u and d should be chosen to match volatility (CRR model). Thus:

$$u = e^{\sigma\sqrt{\Delta t}} \quad \text{and} \quad d = e^{-\sigma\sqrt{\Delta t}}$$

The formulas for matching volatility in real world (with real probabilities) give the same results. *Gyrsanov's theorem*: when we move from the risk-neutral world to the real world, the expected return from the stock price changes but its volatility remains the same.

As the number of time steps is increased (so that Δt becomes smaller), the binomial tree model makes the same assumptions about stock price behavior as the Black–Scholes–Merton model. When the binomial tree is used to price a European option, the price converges to the Black–Scholes–Merton price, as expected, as the number of time steps is increased.

3.) Black-Scholes-Merton Model (BSM)

This is one of the most well-known and widely used methods for pricing European-style options. It assumes constant interest rates and volatility, no dividends, and follows a lognormal distribution of stock prices (Geometric Brownian Motion), efficient markets (no arbitrage and all information are known [Markov property]), and continuous trading. The BSM model provides analytical solutions for option prices.

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \quad (15.20)$$

and

$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \quad (15.21)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

4.) Monte Carlo Simulation

Monte Carlo methods involve generating random simulations of the underlying asset's price movement over time. By running many simulations, the option price can be estimated based on the distribution of potential outcomes. Monte Carlo simulation can accommodate complex features such as path-dependent options and stochastic volatility.

5.) Finite Difference Methods

These methods discretize the Black-Scholes partial differential equation (PDE) governing option prices and solving it numerically. Finite difference methods can handle more complex option structures and boundary conditions compared to the BSM model. ([Investment specialization only](#))

The Greeks – Risk Types

The Greek letters quantify different aspects of the **risk** in an option position. Behind the Greeks we assume BSM model for European option and binomial tree model for American options.

Delta: the rate of change of the option price with respect to the price of the underlying asset. It is the slope of the curve that relates the option price to the underlying asset price.

$$\Delta = \frac{\partial V}{\partial S}$$

It is the *number of units of the stock* we should hold (sell) for each option shorted (bought) in order to create a riskless portfolio. The construction of a riskless portfolio is sometimes referred to as *delta hedging*. The **delta of a call option is positive**, whereas the **delta of a put option is negative**.

Gamma: the rate of change of the option's delta with respect to the price of the underlying asset (second derivative of option with respect to the underlying stock). It measures by how much or how often a position must be rehedge.

$$\Gamma = \frac{\partial^2 V}{\partial S^2}$$

Speed: the rate of change of the option's gamma with respect to the stock price.

$$\text{Speed} = \frac{\partial^3 V}{\partial S^3}$$

Theta: the rate of change of the option's price with respect to time with all else remaining the same. It is sometimes referred to as the time decay of the portfolio. Usually, when theta is quoted time is measured in days, so that theta is the change in the portfolio value when 1 day passes with all else remaining the same.

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$$\Theta = \frac{\partial V}{\partial t}$$

Vega: the rate of change in the option's value with respect to the volatility of the underlying asset (for volatility we use implied volatility). It is a derivative with respect to a parameter, not a variable, and in practice the volatility is not known, it is also hard to predict.

$$\text{Vega} = \frac{\partial V}{\partial \sigma}$$

Rho: the rate of change of the option's price with respect to the interest rate (r). It measures the sensitivity of the value of the portfolio to a change in the interest rate when all else remains the same. In practice r is usually set equal to the risk-free rate for a maturity equal to the option's maturity. This means a trader has exposure to movements in the whole term structure when the options in the trader's portfolio have different maturities.

$$\rho = \frac{\partial V}{\partial r}$$